

KARTIK TRIPATHI

DATA ANALYST & BUSINESS INTELLIGENCE DEVELOPER

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PROFESSIONAL SUMMARY

Analytical and results-driven Data Analyst & BI Developer with a strong quantitative foundation (B.Sc. in Physics, Chemistry, & Mathematics) and professional certification in Data Analytics with GenAI. Proven expertise in SQL database architecture (3NF / Star Schema), Python data analytics pipelines (Pandas, NumPy, SciPy), inferential statistical modeling, and executive dashboard engineering using Power BI and Tableau. Adept at translating multi-million row transactional datasets into actionable commercial insights, predictive customer segmentation (RFM), and interactive business intelligence solutions for cross-functional stakeholders.

CORE COMPETENCIES & TECHNICAL SKILLS

- SQL & Database Engineering: PostgreSQL, MySQL, SQLite, Relational Schema Normalization (3NF), Star Schema, Dimensional Modeling, Window Functions (ROW_NUMBER, RANK), CTEs, Multi-Table Joins, Correlated Subqueries (EXISTS/NOT EXISTS), Indexing & Constraints.
- Python & Data Analytics: Pandas, NumPy, SciPy, Matplotlib, Seaborn, Flask, Jupyter Notebooks, Data Cleansing, ETL Pipelines, Generative AI & Prompt Engineering.
- Business Intelligence & Visualization: Microsoft Power BI (DAX Measures, Power Query ETL, Star Schema Modeling, Interactive Dashboards), Tableau Desktop (Calculated Fields, FIXED LOD Expressions, GIS Mapping), Advanced Excel (Pivot Tables, Dynamic Slicers, XLOOKUP, Nested Formulas, Variance Modeling).
- Statistical Modeling & Commercial Analytics: Inferential Statistics, Hypothesis Testing (Welch's t-Test, ANOVA, Chi-Square, Pearson r), RFM Customer Segmentation, Pareto (80/20) Decile Modeling, KPI Definition, Revenue & Margin Variance Analysis.

FEATURED TECHNICAL PROJECTS

AegisLife Insurance: Risk Analytics & Claims Intelligence (Capstone)

Python (SciPy, Pandas) | SQLite | SQL DDL | Power BI | Inferential Statistics & Fraud Detection

- Relational ETL & Data Ingestion: Ingested, standardized, and validated 2M+ records across 5 transactional files (CustomerMaster, PolicyDetails, ClaimHistory, AgentInfo, CustomerFeedback) into a normalized SQLite database with automated business rule validation.
- Statistical Hypothesis Testing: Executed parametric and non-parametric hypothesis tests in SciPy; disproved legacy risk score predictive validity ($r \approx 0.012$, $p = 0.64$) against actual claim payouts, establishing an evidence-based case for automated ML underwriting.
- Fraud Detection & Anomaly Isolation: Executed Chi-Square tests of independence ($\chi^2 = 38.42$, $p < 0.001$) across policy records to isolate intermediary agent fraud clusters and uncover early-claim anomalies (<90 days post-issuance), addressing a 49% claims suspicion rate.
- Executive BI Dashboard: Engineered an interactive Power BI dashboard tracking loss ratios, claims turnaround cycle times, and geographic policy exposures, delivering an executive roadmap for straight-through processing (STP).

Customer Lifetime Value & RFM Segmentation Analytics Pipeline

Python (Pandas, NumPy, Seaborn) | Jupyter | Behavioral Clustering | Pareto Decile Modeling

- Behavioral Cohort Modeling: Ingested and cleansed 23,050 transactional records across 1,000 unique customer accounts using Pandas, engineering Recency, Frequency, and Monetary (RFM) quintiles via `pd.qcut()` to classify customers into 8 distinct behavioral cohorts.
- Revenue Concentration & Lifetime Value: Evaluated segment financial metrics, revealing that Champions (12.7% of accounts) and Loyal Customers (19.1%) generated a combined 38.4% of total revenue (\$8.85M), establishing target groups for VIP retention.
- Pareto Decile Distribution: Built a dual-axis Pareto distribution model mapping decile spend against cumulative revenue share, confirming top 20% of spenders (Deciles D1–D2) accounted for 27.0% (\$6.23M) and top 70% accounted for 78.3% of total revenue (\$23.05M).
- Commercial Monetization Playbooks: Formulated data-backed retention strategies, including automated win-back workflows for At-Risk accounts (118-day average recency) and cross-sell pathways for high-potential cohorts.

Retail E-Commerce Relational Database & Advanced SQL Analytics

PostgreSQL / MySQL / SQLite | 3NF Schema Architecture | Window Functions & CTEs

- Relational Schema Architecture: Architected and normalized a 6-table relational database (customers, products, orders, order_items, payments, reviews) in 3NF with PK/FK constraints, CHECK rules, and unique indices to ensure transactional integrity.
- Commercial KPI Analytics: Constructed multi-table analytical SQL queries utilizing CTEs, window functions (`ROW_NUMBER() OVER (PARTITION BY ...)`), and aggregate logic to calculate gross revenue, average order value (AOV), SKU velocity, and settlement reconciliations.
- Query & Catalog Optimization: Authored optimized correlated subqueries (NOT EXISTS) to identify dormant accounts and zero-velocity catalog items without resource-intensive outer joins.

Enterprise Business Intelligence, Trend Analysis & Forecasting Dashboards

Tableau Desktop (LOD, GIS) | Power BI (DAX) | Advanced Excel (Variance Modeling)

- Longitudinal Real Estate Analytics: Modeled 12 years of property transaction records (380,000+ entries) in Tableau using FIXED Level of Detail (LOD) expressions and GIS choropleth mapping to isolate pandemic market surges and regional growth trajectories.
- Valuation & Discount Benchmarking: Engineered municipal Sales-to-Assessment ratio KPIs (1.08–1.11 benchmarks) to pinpoint undervalued submarkets transacting at institutional acquisition discounts.
- Operational Margin & Delay Impact Analysis: Modeled commercial flight performance in Power BI using custom DAX measures, demonstrating that operational ground delays exceeding 25 minutes resulted in an average 18.4% drop in operating profit margin.
- Multi-Table Variance Modeling: Built an interactive Amazon e-commerce sales model in Excel with dynamic slicers, XLOOKUP, and variance tracking against regional revenue quotas across 1,000 orders.

PROFESSIONAL EXPERIENCE

Quantitative & STEM Educator (Mathematics & Science) | Self-Employed

Nainital, Uttarakhand, India | January 2021 – Present

- Quantitative Instruction: Delivered structured instruction in Advanced Mathematics, Physics, and Chemistry to 50+ secondary and senior-secondary students, strengthening analytical problem-solving and quantitative reasoning.
- Academic Performance Analytics: Developed Excel-based student performance trackers utilizing lookup functions (XLOOKUP), pivot tables, and statistical formulas to monitor examination scores, pinpoint individual learning gaps, and improve cohort pass averages by 25–30%.
- Performance Reporting & Stakeholder Communication: Synthesized test results, accuracy trends, and milestone progress into clear monthly evaluation reports for parents, communicating actionable feedback and individualized study plans.
- Assessment Design: Created 100+ diagnostic tests, quizzes, and automated scoring rubrics, maintaining a 95%+ student pass and excellence rate across 5 consecutive academic cycles.

CERTIFICATIONS

- Data Analytics with GenAI (Master Certificate): Career 247 (An Adda247 Education Company) | Credential ID: SIN: C2473293
- Generative AI Skills for Business Intelligence (AI0279EN): Career 247 & IBM SkillsNetwork (August 2026)
- AI Applications with Python and Flask (PY0222EN): Career 247 & IBM SkillsNetwork (August 2026)
- Generative AI Essentials (AI0121EN): Career 247 & IBM SkillsNetwork (July 2026)

EDUCATION

Bachelor of Science (B.Sc.) in Physics, Chemistry, and Mathematics

Kumaun University, Nainital, Uttarakhand, India | Graduation Year: 2020